

Hybrid Level-Flow Route to an L^∞ Graph

May 18, 2026

1 The Intended Hybrid

The intended route is not a pure Plan 3 single-level argument and not the full Plan 2 foliation argument. It is the following hybrid.

1. Move inward along torsion levels $E_t = \{u_\Omega > t\}$.
2. Stop at a level $E_{\hat{t}}$ for which the discarded layer is still sharp:

$$|\Omega \setminus E_{\hat{t}}| \leq C_n \delta_T(\Omega).$$

3. Prove that this level is an L^∞ radial graph after translation and volume normalization:

$$E_{\hat{t}}^\# = \{z + r\theta : 0 < r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty(\mathbb{S}^1)} \leq \eta_0.$$

No C^1 , curvature, perimeter, or H^1 control is required.

4. Apply the direct small-amplitude radial-graph theorem:

$$\delta_T(E_{\hat{t}}) \geq c \|\varphi\|_{L^2}^2 \geq c' \mathcal{A}(E_{\hat{t}})^2.$$

5. Transfer back using the superlevel set estimate

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{t}}) + 2|\Omega \setminus E_{\hat{t}}|/|\Omega|.$$

The point of the direct radial-graph theorem is exactly that Step 3 only has to produce a small-amplitude graph. The usual high-norm graph entry is not needed.

2 The Conditional Assembly

The following theorem is already enough to close the hybrid in dimension 2.

Conjecture 2.1 (Level-flow L^∞ graph-entry theorem in $n = 2$). *There are computable constants $\delta_0, \eta_0, C > 0$ such that, if $\Omega \subset \mathbb{R}^2$, $|\Omega| = \pi$, and $\delta_T(\Omega) \leq \delta_0$, then there exists a regular value $\hat{t}$ of u_Ω and a center $z \in \mathbb{R}^2$ such that*

$$|\Omega \setminus E_{\hat{t}}| \leq C \delta_T(\Omega),$$

and, after translating by z and rescaling to area π ,

$$E_{\hat{t}}^\# = \{re^{i\theta} : 0 < r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty(\mathbb{S}^1)} \leq \eta_0,$$

with the volume and first moment normalized.

Proposition 2.2 (Conditional closure of the hybrid route). *Assume Conjecture 2.1. Then the sharp quantitative Saint–Venant estimate*

$$\mathcal{A}(\Omega)^2 \leq C_2 \delta_T(\Omega)$$

holds in dimension 2, with computable constants.

Proof. Choose $E_{\hat{t}}$ from Conjecture 2.1. Step 3 of Plan 3 gives the exact deficit transfer

$$\delta_T(E_{\hat{t}}) \leq \frac{1}{1 - |\Omega \setminus E_{\hat{t}}|/|\Omega|} \delta_T(\Omega) \leq 2\delta_T(\Omega)$$

after reducing δ_0 . The direct radial-graph theorem applies to the normalized graph $E_{\hat{t}}^\sharp$ and gives

$$\mathcal{A}(E_{\hat{t}})^2 = \mathcal{A}(E_{\hat{t}}^\sharp)^2 \leq C \delta_T(E_{\hat{t}}^\sharp) = C \delta_T(E_{\hat{t}}) \leq C \delta_T(\Omega),$$

where the equality uses scale invariance of the normalized deficit. Finally, Step 5 gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{t}}) + 2|\Omega \setminus E_{\hat{t}}|/|\Omega| \leq C\delta_T(\Omega)^{1/2} + C\delta_T(\Omega).$$

Squaring and reducing δ_0 gives the claim. $\square$

3 What Is Already Proved

- The $O(\delta_T)$ layer selection is proved in Plan 3 Step 1, using the volume variable $s = m(t)$. This avoids the old t -length gap.
- The deficit transfer to $E_{\hat{t}}$ is proved in Plan 3 Step 3, using the exact identity $u_\Omega - \hat{t} = u_{E_{\hat{t}}}$.
- The transfer back to Ω is proved in Plan 3 Step 5.
- The small-amplitude radial-graph theorem is proved in `H12_ASYMMETRY_DIRECT_PROOF.tex`. It needs only $\|\varphi\|_{L^\infty}$ small and $\|\varphi\|_{L^2}$ small; it does not require a slope or curvature bound.

4 What the Entry Theorem Must Prove

Conjecture 2.1 is stronger than mere annular containment. It must prove both:

$$B_{1-\eta_0}(z) \subset E_{\hat{t}}^\sharp \subset B_{1+\eta_0}(z),$$

and one radial crossing from z in every direction. These are separate issues.

4.1 Outward fingers

The ball-plus-tube model supports the plan in dimension 2. For a tube of width w and length $L = O(1)$,

$$\delta_T(\Omega) \simeq wL,$$

while clearing the tube by passing above its cap height costs a ball-shell volume of order w^2 , plus the tube volume wL . Since $w^2 \leq CwL$ in the nontrivial regime, the tube can be cleared within an $O(\delta_T)$ discarded layer in $n = 2$. Also, on active tube levels,

$$D_H(t) \simeq L/w,$$

so those levels are very expensive for the D_H -identity. This is the main reason the $n = 2$ hybrid remains plausible.

4.2 Inward fjords and holes

The harder part is inward geometry. Since $E_t \subset \Omega$, moving inward cannot fill a missing fjord or hole. Therefore the entry theorem must show that a small-deficit planar domain cannot have a fjord penetrating a fixed distance into the ball, unless that fjord is already paid for by $\delta_T(\Omega)$ at a fixed scale.

The natural tool is planar capacity/modulus. A narrow slit of fixed depth has positive $W^{1,2}$ -capacity in the disk, so imposing $u = 0$ on such a slit should lower torsional rigidity by a definite amount, independently of the slit width. A quantitative version would imply:

$$\text{inward penetration depth} \geq \eta \implies \delta_T(\Omega) \geq c(\eta) > 0.$$

This would give the inner-radius half of the L^∞ graph entry after choosing the small-deficit threshold below $c(\eta_0)$. This capacity lemma is not currently proved in the repo.

4.3 Single crossing

Annular containment still permits multiple crossings on a ray. To feed the radial-graph theorem one must rule out those crossings, or replace the graph theorem by a general annular-set theorem.

For the hybrid as stated, the required lemma is:

multiple crossings before the $O(\delta_T)$ stop $\implies$ either large D_H on a positive level interval or fixed torsion

Plan 2's good-ray/bad-ray slicing machinery is relevant here: bad rays are measured by multiplicity and can be absorbed by perimeter-type excesses. But the current repo does not contain the torsion-energy lower bound that turns this multiplicity into the L^∞ graph-entry conclusion.

5 What Not To Use

The hybrid should not use the single-level Hessian Chain B. The false step is

$$\delta_T(D) \gtrsim \int_D |D^2 u_D + \tfrac{1}{n} I|^2,$$

which fails on high-frequency smooth near-balls by the spectral calculation recorded in `PLAN3_N2_SINGLE_LEVEL`. The correct scale is the radial-graph theorem's L^2 /Fraenkel scale, not an H^1 Hessian scale.

6 Precise Status

The hybrid route is coherent and sharper than the previous annular-set formulation: if the stopped level is truly an L^∞ radial graph, the existing radial-graph theorem finishes the proof. The missing theorem is exactly Conjecture 2.1.

The proof of that conjecture should likely split into:

1. a D_H -integral clearing lemma for outward fingers, using the $n = 2$ scaling $D_H \simeq L/w$ on active tube levels;
2. a planar capacity/modulus exclusion of inward fjords of fixed depth;
3. a same-center bad-ray/multiplicity argument to promote annular containment to one radial crossing.

None of these three ingredients is currently written as a proof in the repo. The model computations support them in $n = 2$, but they are the actual work left by the hybrid plan.